Uniform Coefficient Transseries across the Polylogarithmic Order-Three Resonance

Exact normalization, Gaussian order derivatives,
Lambert transitions, and reciprocal-order inverse germs

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Abstract

The quadratic-feedback equation $uv - v^2/2 = \mathrm{Li}_s(re^v)$ has different coefficient laws below, at, and above $s = 3$. Recent repository work determines its fixed-order laws and a joint varying-order inverse chart, but explicitly leaves the corresponding joint coefficient problem open. We resolve that problem when $(s-3)\log n$ remains bounded, allowing the feedback parameter to vary simultaneously. A pole-paired scalar equation defines an exact Gaussian scale. In that scale every inverse-logarithmic correction is a finite Stirling-number combination of order derivatives of a parabolic cylinder function. We prove an absolute remainder of order $(\log n)^{-N-1}$ at every fixed order, uniformly over all lattice locations and compact parameter sets.

At exact resonance the scale is expressed by the lower Lambert branch. At the center the correction functions reduce to finite Gamma moments, and reproduce the previous marginal coefficient polynomials at every order. More generally the entire logarithmic expansion survives positive analytic composition of the polylogarithm; the nonlinear composition enters only through its first two endpoint derivatives after centering and endpoint normalization. The same expansion yields relative positive-tail matching in every fixed power-of-$\log n$ window and an explicit Lambert location for the Gaussian–power-tail transition.

A separate local theorem classifies finite ramification of inverse polylogarithms for every real $s > 1$. The endpoint is removable precisely at $s = 1 + 1/m$, $m \geq 2$. We give a convergent crossover chart through those orders and prove that the integer cut-modulus argument cannot extend to $1 < s < 3/2$. Complete analytic proofs, a focused source audit, exact symbolic checks, numerical diagnostics, and further research targets accompany the report.

Scope and status. This is an unrefereed research manuscript with conventional proofs and reproducible checks. The results are not proof-assistant formalized. The contribution is specified relative to the inspected repository snapshot and the primary literature; worldwide publication priority is not asserted. All coefficient expansions are fixed-order asymptotic statements. The reciprocal-order local chart has a separate, proved convergence assertion.

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1 The research target and its antecedents

1.1 An explicit unresolved coefficient problem

The inspected repository is pinned to commit

`8f5afd4d7565df808359f6a0c22f87f79a119bb9`.

The user-specified Polylogarithms and Transseries directories and all three archives present in `docs/incoming` at that commit were inventoried. The relevant integrated chapters and predecessor reports were read directly. This is a focused audit of the claims used here, not a review of the entire repository.

Question Q1 in the boundary-completion report [1] asks for a coefficient theorem with

$$L = \log n, \qquad s = 3 + \varepsilon, \qquad \lambda = \frac{\varepsilon L}{2} \tag{1.1}$$

bounded, while the feedback distance from criticality varies on its natural local-limit scale. That report proves a convergent joint *inverse* chart but keeps s fixed in its coefficient theorem. Older linear-tail reports provide leading moving-order limits and a first central correction under eventually exact power-tail hypotheses [2, 3]. The quadratic outer function changes infinitely many Taylor coefficients, so those hypotheses do not literally include the present model. At exactly $s = 3$, all-order logarithmic corrections already appear in the marginal reports and the integrated Gamma-moment chapter [4, 5].

We prove the missing joint theorem with every fixed logarithmic order, including noncentral locations. Its Fourier remainder is uniform over the whole lattice. A further argument converts that absolute control into a relative result in a specified expanding positive window. The latter is a partial positive-side resolution of Q2 in [1]; it does not cover arbitrary deviations or the negative saddle window.

1.2 What comes from classical analysis

The convergent Jonquière expansion, Gamma recurrence, and polylogarithm continuation are classical [7]. Lagrange inversion and its probability interpretation supply the exact coefficient representation already used in the repository. Neither these tools nor Gaussian local limits are claimed as discoveries.

There is also substantial prior work on nonclassical corrections to Gaussian limits. Lam, Blanchet, Burch and Bazant [10] express heavy-tail corrections using parabolic cylinder functions and discuss logarithmic effects and matching to the tails. Berger, Birkner and Yuan [11] study Gaussian and single-large-jump local deviations, including the critical tail index two. Thus the appearance of parabolic cylinder functions and the Gaussian–power-tail transition are established phenomena. Their occurrence here is useful because the polylogarithmic order moves with n through the collision of two singular terms, and the nonlinear feedback is retained exactly.

The proposed contribution is the combined formulation: a uniformly pole-paired scale, a finite all-order Stirling calculus for a moving order and moving feedback, a fully proved uniform remainder, and the two-derivative composition principle. The local ramification theorem later in the article is another direct deduction from the classical polylogarithm expansion, with a precise repository application. The limited comparison performed here cannot certify first priority in the entire literature.

1.3 Results at a glance

Result	Mathematical content
Exact normalization	A unique scale $B > 1$ solves a paired Gamma–zeta equation; it is a Lambert expression when $s = 3$.
All fixed orders	A finite sum of Gaussian order derivatives gives error $O(L^{-N-1})$, uniform in lattice location.
Central identities	Every central correction is a finite Gamma moment; the marginal coefficients match the previous ones exactly.
Composition principle	Higher endpoint derivatives of an analytic outer function enter only beyond all fixed inverse-logarithmic orders.
Expanding positive window	Relative power-tail matching holds up to any fixed power of L; the transition has an explicit Lambert center.
Inverse endpoint theorem	Minimal finite ramification is classified for every real $s > 1$, including all removable reciprocal orders.

1.4 Notation and the meaning of the asymptotics

All moving-order theorems concern real ε with $|\lambda| \le \Lambda$ for an arbitrary fixed $\Lambda < \infty$. The positive feedback coordinate b remains in a fixed compact interval $[b_-, b_+] \subset (0, \infty)$. Constants in an O term may depend on these sets and the fixed truncation order, but not on n, the parameter choice within them, or the lattice location when uniformity in that location is asserted. A statement to every fixed order does not permit the truncation order to grow with n.

The logarithm $\operatorname{Log} t$ is principal on the slit plane. On the Fourier axis,

$$\operatorname{Log}(-iy) = \log|y| - \frac{\pi i}{2}\operatorname{sgn}(y) \qquad (y \in \mathbb{R} \setminus \{0\}). \tag{1.2}$$

The normal density is $\mathfrak{g}(\xi) = (2\pi)^{-1/2} e^{-\xi^2/2}$. The notation $\mathcal{H}_a^{[j]}$ always means differentiation in the *order* a, not in the spatial variable ξ. The coefficients $\left\{ {j \atop k} \right\}$ are Stirling numbers of the second kind.

Inverse-logarithmic sectors in L are transseries sectors in the usual asymptotic sense. The nonlinear local remainder below is of size n^{-c} times logarithms, hence of exponential size e^{-cL} relative to those sectors. We identify its scale separation but do not claim to compute all such exponential sectors.

2 The exact coefficient law and the resonant scale

2.1 Quadratic feedback as a compound-Poisson coefficient

For real $s > 2$ and $b > 0$, set

$$u = \sqrt{b^2 + 2\zeta(s)}, \qquad A = \zeta(s-1), \qquad v_c = u - b, \qquad R_b = e^{-v_c}. \tag{2.1}$$

Consider the unique germ

$$uv(r) - \frac{v(r)^2}{2} = \operatorname{Li}_s(re^{v(r)}), \qquad v(0) = 0, \qquad a_n = [r^n]v(r). \tag{2.2}$$

Define

$$f(z) = u - \sqrt{u^2 - 2\,\mathrm{Li}_s(z)} = \sum_{k\geq 1} c_k z^k, \tag{2.3}$$

using the square root equal to u at zero. Since $2\zeta(s) < u^2$, the Taylor expansion of the outer square root converges absolutely on the whole image of $|z| \leq 1$. In particular,

$$c_k \geq \frac{1}{uk^s} > 0, \qquad \sum_{k\geq 1} c_k = f(1) = v_c, \qquad f'(1) = \frac{A}{b}. \tag{2.4}$$

Proposition 2.1 (Exact coefficient–probability identity)**.** *Let X_n be the nonnegative integer-valued compound-Poisson variable with probability generating function*

$$\mathbb{E}z^{X_n} = \exp\{n(f(z) - f(1))\}. \tag{2.5}$$

Then

$$R_b^n a_n = \frac{1}{n}\mathbb{P}(X_n = n), \qquad \mathbb{E}X_n = \frac{nA}{b}. \tag{2.6}$$

The identity holds for every $b > 0$, including $b < A$.

Proof. Put $z = re^v$ in (2.2). Solving the quadratic equation on its germ gives $v = f(z)$, so $z = re^{f(z)}$. Lagrange inversion gives

$$a_n = \frac{1}{n}[z^{n-1}]f'(z)e^{nf(z)} = \frac{1}{n}[z^n]e^{nf(z)}.$$

For the second equality differentiate $e^{nf(z)}$ and compare its z^{n-1} coefficient. Positivity and summability in (2.4) make $\exp\{n\sum c_k(z^k - 1)\}$ a probability generating function: take a Poisson number of jumps of total intensity $nf(1)$, with jump probabilities $c_k/f(1)$. Multiplication by $e^{-nf(1)}$ proves the coefficient identity. Differentiation at $z = 1$ proves the mean formula. □

The endpoint reference R_b is the actual Taylor radius at and above $b = A$. Below that threshold the branch first encounters an interior fold and the radius is larger than R_b [1]. No radius claim is needed for (2.6).

2.2 Pairing the colliding Gamma and zeta terms

Write $s = 3 + \varepsilon$ and define

$$E_2(\varepsilon) = \frac{\Gamma(1-\varepsilon)}{(1+\varepsilon)(1+\varepsilon/2)}, \qquad Z(\varepsilon) = \zeta(1+\varepsilon) - \varepsilon^{-1}, \tag{2.7}$$

$$K_\varepsilon(\ell) = Z(\varepsilon) + \frac{1 - E_2(\varepsilon)e^{-\varepsilon\ell}}{\varepsilon}. \tag{2.8}$$

Both expressions are interpreted by removal of the singularity at $\varepsilon = 0$. The elementary values are

$$E_2(0) = 1, \quad E_2'(0) = \gamma - \frac{3}{2}, \quad Z(0) = \gamma, \quad K_0(\ell) = \ell + \frac{3}{2}. \tag{2.9}$$

Gamma recurrence gives $2\Gamma(-2-\varepsilon) = -E_2(\varepsilon)/\varepsilon$. Pairing that term with the zeta pole in the convergent Jonquière expansion yields the following uniform local statement.

Lemma 2.2 (The centered local exponent)**.** *There are $r > 0$ and $C < \infty$, uniform for $b \in [b_-, b_+]$ and real $|\varepsilon| \le 1/4$, such that, for $0 < |t| < r$ and $|\arg t| \le \pi/2$,*

$$f(e^{-t}) - f(1) + \frac{A}{b}t = \frac{t^2}{2b}\{K_\varepsilon(-\operatorname{Log} t) + \nu\} + \mathcal{E}(t), \qquad \nu = \frac{A^2}{b^2}, \tag{2.10}$$

where $|\mathcal{E}(t)| \le C|t|^{5/2}$.

Proof. The classical normally convergent expansion [7, 25.12.12] gives, after pairing,

$$D_s(t) := \zeta(s) - \operatorname{Li}_s(e^{-t}) = At - \frac{t^2}{2}K_\varepsilon(-\operatorname{Log} t) + t^3 J_\varepsilon(t), \tag{2.11}$$

where J_ε is jointly holomorphic for t near zero and ε near the indicated compact interval. Normal convergence can also be checked by the zeta functional equation: the coefficients of the analytic tail have geometric majorants on each smaller disk $|t| < r < 2\pi$. All poles near $\varepsilon = 0$ occur in the displayed paired term.

Using

$$\frac{1 - e^{-\varepsilon\ell}}{\varepsilon} = \ell \int_0^1 e^{-\varepsilon\ell v}\, dv$$

and the bounded regular part of $(E_2(\varepsilon) - 1)/\varepsilon$ gives

$$|K_\varepsilon(-\operatorname{Log} t)| \le C|t|^{-1/4}(1 + |\log|t||) \le C'|t|^{-1/2}.$$

Thus $D_s(t) = At + O(|t|^{3/2}) = O(|t|)$. Uniformly in b,

$$b - \sqrt{b^2 + 2D} = -\frac{D}{b} + \frac{D^2}{2b^3} + O(|D|^3).$$

Inserting (2.11) leaves the displayed terms of (2.10); the difference between D^2 and A^2t^2 is $O(|t|^{5/2})$. This proves the bound, including the imaginary boundary rays by continuation and continuity. □

The nonlinear feedback contributes precisely ν to the quadratic regular part. At criticality $b = A$, this is 1, changing the resonant constant from 3/2 to 5/2.

2.3 A scalar equation absorbing the slow logarithms

Define $B = B(n, \varepsilon, b) > 1$ by

$$\boxed{V := \frac{bB^2}{n} = K_\varepsilon(\log B) + \nu.} \tag{2.12}$$

This defines a scale from the paired quadratic exponent; it does not assume that the underlying distribution has a finite variance. Put

$$q = \frac{E_2(\varepsilon)B^{-\varepsilon}}{V}, \qquad \xi_m = \frac{m - nA/b}{B}, \qquad \xi = \xi_n. \tag{2.13}$$

Theorem 2.3 (Existence, uniqueness, and size of the scale)**.** *Uniformly for* $|\lambda| \le \Lambda$ *and* $b \in [b_-, b_+]$, (2.12) *has exactly one solution* $B > 1$ *for all sufficiently large* n*. Let*

$$\varphi(\lambda) = \frac{1 - e^{-\lambda}}{\lambda}, \quad \varphi(0) = 1, \qquad B_0^2 = \frac{nL}{2b}\varphi(\lambda). \tag{2.14}$$

Then

$$\frac{B}{B_0} = 1 + \frac{C_\lambda}{L\varphi(\lambda)} + O\left(\frac{(\log L)^2}{L^2}\right), \tag{2.15}$$

$$C_\lambda = \gamma + \nu + e^{-\lambda}\left\{\frac{1}{2}\log L + \frac{1}{2}\log\frac{\varphi(\lambda)}{2b} - \gamma + \frac{3}{2}\right\}, \tag{2.16}$$

$$q = \frac{2e^{-\lambda}}{L\varphi(\lambda)} + O\left(\frac{\log L}{L^2}\right). \tag{2.17}$$

In particular $q > 0$ *and* $q \asymp L^{-1}$ *on the specified parameter sets.*

Proof. Set $H = \log B$ and $F(H) = be^{2H}/n - K_\varepsilon(H) - \nu$. For small real ε, $K_\varepsilon(0) + \nu$ is uniformly positive by (2.9). Thus $F(0) < 0$ for large n. Moreover

$$F'(H) = e^{-\varepsilon H}\left\{\frac{2b}{n}e^{(2+\varepsilon)H} - E_2(\varepsilon)\right\}.$$

The braced expression increases strictly, because $2 + \varepsilon > 0$. It follows that F first decreases and then increases, or increases throughout. In both cases $F(H) \to +\infty$ as $H \to \infty$: the growth exponent 2 dominates the possible growth exponent $-\varepsilon$ of K_ε. Hence there is exactly one positive zero.

Write $H_0 = \log B_0 = \frac{1}{2}\{L + \log L + \log(\varphi(\lambda)/(2b))\}$. The representation (2.8), with $\varepsilon = 2\lambda/L$, gives uniformly, including $\lambda = 0$,

$$K_\varepsilon(H_0) + \nu = \frac{L}{2}\varphi(\lambda) + C_\lambda + O((\log L)^2/L).$$

For example the first term follows from $(1 - e^{-\lambda})/\varepsilon = L\varphi(\lambda)/2$; the next term uses $E_2'(0) = \gamma - 3/2$. First bracket the root by $B = cB_0$ for fixed constants $c < 1$ and $c > 1$; the leading terms have opposite signs. Then the derivative at the bracketed root is of order L. Taylor's theorem in $H - H_0$ improves the bracket to $H - H_0 = C_\lambda/(L\varphi(\lambda)) + O((\log L)^2/L^2)$. Exponentiation gives (2.15). Finally $E_2(\varepsilon)B^{-\varepsilon} = e^{-\lambda}(1 + O(\log L/L))$ and $V = L\varphi(\lambda)/2 + O(\log L)$, proving the q formula. □

3 Gaussian order derivatives and finite correction formulas

For $\operatorname{Re} a > -1$ and real ξ, define

$$\mathcal{H}_a(\xi) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-i\xi y - y^2/2}(-iy)^a\, dy, \qquad \mathcal{H}_a^{[j]}(\xi) = \partial_a^j \mathcal{H}_a(\xi). \tag{3.1}$$

The logarithm convention is (1.2). All fixed order derivatives are legitimate by domination at zero and infinity. For real a and ξ the two halves of the integral are conjugates.

Proposition 3.1 (Parabolic cylinder and Gamma forms)**.** *In the standard parabolic-cylinder normalization,*

$$\mathcal{H}_a(\xi) = \frac{e^{-\xi^2/4}}{\sqrt{2\pi}} D_a(-\xi), \tag{3.2}$$

$$\mathcal{H}_a(0) = \frac{2^{(a-1)/2}}{\Gamma((1-a)/2)}, \qquad \partial_\xi \mathcal{H}_a = \mathcal{H}_{a+1}. \tag{3.3}$$

In particular $\mathcal{H}_0 = \mathfrak{g}$ *and* $\mathcal{H}_{2k} = \mathfrak{g}^{(2k)}$*. Each function* $\mathcal{H}_a^{[j]}$ *used below is an ordinary convergent integral, even when a later asymptotic series containing it does not converge.*

Proof. The derivative identity follows immediately under the integral. Splitting into $y > 0$ and $y < 0$, then using the Gamma integral, gives

$$\mathcal{H}_a(0) = \frac{2^{(a-1)/2}}{\pi} \Gamma((a+1)/2) \cos(\pi a/2).$$

The reflection formula gives (3.3). Integration by parts in y, with vanishing boundary term $y(-iy)^a e^{-y^2/2 - i\xi y}$, gives

$$\mathcal{H}_a'' + \xi \mathcal{H}_a' + (a+1)\mathcal{H}_a = 0.$$

The function on the right of (3.2) satisfies the same equation by the parabolic-cylinder differential equation. The values of $D_a(0)$ and $D_a'(0)$ [8, 12.2.4–12.2.7] give the same value and first derivative at $\xi = 0$, namely $2^{(a-1)/2}/\Gamma((1-a)/2)$ and $2^{a/2}/\Gamma(-a/2)$. Uniqueness for the ordinary differential equation proves the identity. □

3.1 The correction polynomials in two small parameters

Define the finite functions

$$\mathcal{C}_0(\xi; q, \varepsilon) = \mathfrak{g}(\xi),$$

$$\mathcal{C}_j(\xi; q, \varepsilon) = \frac{1}{j!} \sum_{k=1}^{j} \left\{ {j \atop k} \right\} \left(-\frac{q}{2}\right)^k \varepsilon^{j-k} \mathcal{H}_{2k}^{[j]}(\xi), \qquad j \geq 1. \tag{3.4}$$

They are homogeneous of degree j in (q, ε). The first three are

$$\mathcal{C}_1 = -\frac{q}{2} \mathcal{H}_2^{[1]}, \tag{3.5}$$

$$\mathcal{C}_2 = -\frac{q\varepsilon}{4} \mathcal{H}_2^{[2]} + \frac{q^2}{8} \mathcal{H}_4^{[2]}, \tag{3.6}$$

$$\mathcal{C}_3 = -\frac{q\varepsilon^2}{12} \mathcal{H}_2^{[3]} + \frac{q^2\varepsilon}{8} \mathcal{H}_4^{[3]} - \frac{q^3}{48} \mathcal{H}_6^{[3]}. \tag{3.7}$$

The spatial argument ξ is suppressed on the right. These expressions involve order derivatives of the Gaussian kernel; in general they are not a Gaussian times an ordinary polynomial in ξ.

There is also a finite divided-difference form. For a holomorphic function of a, put

$$\mathfrak{D}_\varepsilon^k F(a) = \frac{1}{\varepsilon^k} \sum_{h=0}^{k} (-1)^{k-h} \binom{k}{h} F(a + h\varepsilon), \qquad \mathfrak{D}_0^k F = F^{(k)}. \tag{3.8}$$

The apparent poles are removable. The identity

$$\left(\frac{e^{\varepsilon z}-1}{\varepsilon}\right)^k = k!\sum_{j\ge k}\left\{{j \atop k}\right\}\frac{\varepsilon^{j-k}z^j}{j!} \tag{3.9}$$

explains the appearance of the Stirling numbers. We will prove that both the q-ordered finite expression

$$\sum_{k=0}^{N}\frac{(-q/2)^k}{k!}\,\mathfrak{D}_\varepsilon^k\mathcal{H}_a(\xi)\Big|_{a=2k} \tag{3.10}$$

and the total-degree expression $\sum_{j=0}^{N}\mathcal{C}_j(\xi;q,\varepsilon)$ approximate the same lattice law with error $O(L^{-N-1})$. They differ from one another only at the omitted orders. Formula (3.10) can be useful for computation away from $\varepsilon=0$; near zero it is better evaluated using derivatives or the removable integral formula.

3.2 Finite Gamma moments at the center

For integers $j,k\ge 1$ set

$$M_j(k) = \operatorname{Re}\left[(\partial_z+\log 2+i\pi)^j\Gamma(z)\right]_{z=k+1/2}. \tag{3.11}$$

This is also the convergent moment

$$M_j(k) = \operatorname{Re}\int_0^\infty e^{-t}t^{k-1/2}(\log t+\log 2+i\pi)^j\,dt. \tag{3.12}$$

Corollary 3.2 (Central coefficient calculus)**.** *For every $j\ge 1$,*

$$\frac{\mathcal{C}_j(0;q,\varepsilon)}{\mathfrak{g}(0)} = \frac{1}{2^j j!\sqrt{\pi}}\sum_{k=1}^{j}\left\{{j \atop k}\right\}q^k\varepsilon^{j-k}M_j(k). \tag{3.13}$$

In particular

$$\frac{\mathcal{C}_1(0;q,\varepsilon)}{\mathfrak{g}(0)} = \frac{q}{4}(2-\gamma-\log 2). \tag{3.14}$$

Proof. Differentiate the Gamma–cosine expression in the proof of proposition 3.1. At $a=2k$ it gives

$$\mathcal{H}_{2k}^{[j]}(0) = \frac{(-1)^k 2^{k-j-1/2}}{\pi}M_j(k).$$

Substitution in (3.4) cancels the two factors $(-1)^k$ and proves (3.13). For $j=k=1$, use $\Gamma(3/2)=\sqrt{\pi}/2$ and $\psi(3/2)=2-\gamma-2\log 2$. □

These formulas can be converted into polygamma values, harmonic numbers, zeta values, and the Euler constant by repeated differentiation of Γ and the half-integer polygamma recurrences. The moment form keeps all signs and factors of two in a single finite expression.

4 The uniform coefficient transseries

We now prove the main result. The definition of B in (2.12) incorporates all logarithmic changes of the quadratic coefficient, including the slowly varying feedback contribution.

Theorem 4.1 (Every fixed logarithmic order, uniformly on the lattice)**.** *Fix $\Lambda < \infty$, $0 < b_- < b_+ < \infty$, and an integer $N \geq 0$. For $L = \log n$, $|\varepsilon L/2| \leq \Lambda$, and $b \in [b_-, b_+]$, let X_n, B, q, and ξ_m be defined by (2.5), (2.12), and (2.13). Then*

$$\sup_{m\in\mathbb{Z}} \left| B\mathbb{P}(X_n = m) - \sum_{j=0}^{N} \mathcal{C}_j(\xi_m; q, \varepsilon) \right| \leq C_N L^{-N-1} \tag{4.1}$$

for all sufficiently large n, uniformly in the stated parameters. Consequently the feedback coefficients satisfy

$$\boxed{nBR_b^n a_n = \mathfrak{g}(\xi) + \sum_{j=1}^{N} \frac{1}{j!} \sum_{k=1}^{j} \left\{ {j \atop k} \right\} \left(-\frac{q}{2}\right)^k \varepsilon^{j-k} \mathcal{H}_{2k}^{[j]}(\xi) + O(L^{-N-1}).} \tag{4.2}$$

The finite divided-difference expression (3.10) has the same error bound.

The theorem resolves the joint coefficient target Q1 in [1]. It is an absolute estimate. For bounded ξ it gives relative Gaussian asymptotics because $\mathfrak{g}$ has a positive minimum on each compact real interval. Relative conclusions for unbounded ξ will be proved separately.

4.1 Uniform control of the whole Fourier contour

Lemma 4.2 (Uniform Fourier domination)**.** *Let*

$$\chi_n(y) = \mathbb{E} \exp\left(iy \frac{X_n - nA/b}{B} \right). \tag{4.3}$$

There are positive constants c_1, c_2, uniform in the moving-order window and in b, such that

$$|\chi_n(y)| \leq e^{-c_1 y^2} \qquad (|y| \leq Bn^{-1/4}), \tag{4.4}$$

$$|\chi_n(y)| \leq e^{-c_2\sqrt{n}} \qquad (Bn^{-1/4} \leq |y| \leq \pi B). \tag{4.5}$$

Proof. By positivity of the intensities,

$$-\log|\chi_n(B\theta)| = n \sum_{k\geq 1} c_k (1 - \cos k\theta).$$

For $|\theta| \leq n^{-1/4}$ retain $1 \leq k \leq \lfloor n^{1/4} \rfloor$. Then $|k\theta| \leq 1$, so $1 - \cos k\theta \geq ck^2\theta^2$. The parameters keep u bounded above. Moreover $k^{-\varepsilon} \geq e^{-\Lambda/2}$ on this range, since $|\varepsilon| \log k \leq \Lambda/2$. Therefore

$$-\log|\chi_n(B\theta)| \geq cn\theta^2 \sum_{k\leq n^{1/4}} k^{-1-\varepsilon} \geq c' nL\theta^2.$$

Since $B^2 \asymp nL$, this proves (4.4). For $n^{-1/4} \leq |\theta| \leq \pi$, retain only $k = 1$. Here $c_1 = 1/u$ in the intensity notation and $1 - \cos\theta \geq 2\theta^2/\pi^2$, giving $-\log|\chi_n(B\theta)| \geq cn\theta^2 \geq c\sqrt{n}$. The centering phase has modulus one, proving the second assertion. □

This argument is insensitive to the cancellation of the Gamma and zeta poles. It uses the exact nonnegative coefficients and supplies the uniformity that a fixed-order local-limit theorem alone would not provide.

4.2 The exact central reduction

For $z \in \mathbb{C}$, define the entire divided exponential

$$h_\varepsilon(z) = \frac{e^{\varepsilon z} - 1}{\varepsilon}, \qquad h_0(z) = z. \tag{4.6}$$

Take $t = -iy/B$ in lemma 2.2. The scale equation removes the constant quadratic term and gives

$$\chi_n(y) = \exp\left\{-\frac{y^2}{2} + \frac{q}{2} y^2 h_\varepsilon(\mathrm{Log}(-iy)) + R_n(y)\right\}, \quad |R_n(y)| \le CnB^{-5/2}|y|^{5/2}, \tag{4.7}$$

whenever $0 < |y| < rB$. The limit at zero is understood continuously. The sign in this formula follows from $(-iy)^2 = -y^2$ and

$$K_\varepsilon(\log B - \mathrm{Log}(-iy)) - K_\varepsilon(\log B) = -E_2(\varepsilon) B^{-\varepsilon} h_\varepsilon(\mathrm{Log}(-iy)).$$

Since $nB^{-5/2} = O(n^{-1/4}L^{-5/4})$, the local nonlinear remainder is smaller than every fixed inverse power of L.

4.3 A finite expansion with an integrable remainder

Proof of theorem 4.1. Choose $T_L = \sqrt{D_N \log L}$ with a sufficiently large constant D_N. By lemma 4.2, the part of the scaled Fourier integral with $T_L < |y| \le \pi B$ has absolute size $O(L^{-N-1})$: use the Gaussian bound up to $Bn^{-1/4}$ and the bound $2\pi B e^{-c_2\sqrt{n}}$ beyond it.

On $|y| \le T_L$, put $z = \mathrm{Log}(-iy)$ and $w = (q/2)y^2 h_\varepsilon(z)$. The inequality

$$|h_\varepsilon(z)| \le |z| e^{|\varepsilon z|} \tag{4.8}$$

follows from $h_\varepsilon(z) = z\int_0^1 e^{v\varepsilon z}\,dv$. It proves $\sup_{|y|\le T_L}|w| = o(1)$. Near $y = 0$ one uses $y^2|\log|y||\,|y|^{-|\varepsilon|} \to 0$; no uniform bound on $|\varepsilon \log|y||$ is assumed. For $1 \le |y| \le T_L$ the bound is $O(L^{-1}\log L \log\log L)$. Equation (4.7) and the estimate on R_n therefore allow us to replace χ_n by $e^{-y^2/2}e^w$ in the central integral with error $O(nB^{-5/2})$.

Expand the exponential only to the finite order N:

$$e^w = \sum_{k=0}^{N} \frac{w^k}{k!} + O_N(|w|^{N+1}).$$

The error is integrable against $e^{-y^2/2}$ with bound $C_N q^{N+1}$. Indeed (4.8) bounds the integrand near zero by a constant times $|y|^{2N+2-(N+1)|\varepsilon|}(1 + |\log|y||)^{N+1}$; at infinity a Gaussian dominates the polynomial and logarithmic factors. For the fixed N, take n large enough that $(N+1)|\varepsilon| < 1/2$. All these integrals then have a common majorant.

For each $k \le N$, use

$$h_\varepsilon(z)^k = z^k \int_{[0,1]^k} e^{\varepsilon z(v_1+\cdots+v_k)}\,dv_1\cdots dv_k. \tag{4.9}$$

Taylor expansion in ε through degree $N-k$ gives

$$h_\varepsilon(z)^k = k! \sum_{j=k}^{N} \left\{ {j \atop k} \right\} \frac{\varepsilon^{j-k} z^j}{j!} + \mathcal{R}_{k,N}(z),$$

where

$$|\mathcal{R}_{k,N}(z)| \le C_N|\varepsilon|^{N+1-k}|z|^{N+1}e^{k|\varepsilon z|}.$$

After multiplication by $q^k|y|^{2k}e^{-y^2/2}$, its integral is $O(q^k|\varepsilon|^{N+1-k}) = O(L^{-N-1})$, by the same domination. Thus the total-degree polynomial in q, ε has an $L^1(dy)$ error of the asserted order. Its finite terms may be extended from $|y| \le T_L$ to the full real axis: their Gaussian tails are $O(L^{-N-1})$ after enlarging D_N if needed.

The resulting integrals are

$$\frac{1}{2\pi}\int_{\mathbb{R}} e^{-i\xi_m y - y^2/2}(-iy)^{2k}(\mathrm{Log}(-iy))^j\,dy = \mathcal{H}_{2k}^{[j]}(\xi_m).$$

Here the factor y^{2k} in w^k contributes $(-1)^k$, producing $(-q/2)^k$. Lattice Fourier inversion says

$$B\mathbb{P}(X_n = m) = \frac{1}{2\pi}\int_{-\pi B}^{\pi B} e^{-i\xi_m y}\chi_n(y)\,dy.$$

Every error bound above is an L^1 bound before insertion of the phase. Since $|e^{-i\xi_m y}| = 1$, it is uniform in all integers m. This proves (4.1). The exact coefficient identity gives (4.2).

If one retains $h_\varepsilon(z)^k$ instead of Taylor-expanding in ε, apply finite Fourier inversion to

$$(-iy)^{2k}h_\varepsilon(\mathrm{Log}(-iy))^k.$$

This gives precisely (3.10). The same q^{N+1} estimate proves its assertion. □

Remark 4.3 (The full formal exponential is not the Fourier law). The proof integrates a finite expansion in the central region and uses the exact characteristic function on its complement. It does not replace the full Fourier law by $\exp\{-y^2/2 + (q/2)y^2h_\varepsilon(\mathrm{Log}(-iy))\}$ on the whole real axis. For fixed positive q, ε the latter expression can grow at large $|y|$. Nor does the theorem assert convergence of the infinite sum over j. The finite integrals and all fixed-order remainders are the claimed objects.

4.4 The elementary normalization requested in Q1

Put $\xi_0 = n(1 - A/b)/B_0$, using (2.14). On compact ξ_0 sets, theorems 2.3 and 4.1 give

$$\begin{aligned} nB_0R_b^n a_n = \mathfrak{g}(\xi_0) + \frac{C_\lambda\mathcal{H}_2(\xi_0) - e^{-\lambda}\mathcal{H}_2^{[1]}(\xi_0)}{L\varphi(\lambda)} \\ + O\left(\frac{(\log L)^2}{L^2}\right). \end{aligned} \tag{4.10}$$

To verify the scale-change term, write $B/B_0 = 1 + \delta$; then $(B_0/B)\mathfrak{g}(\xi_0B_0/B) = \mathfrak{g}(\xi_0) - \delta(\mathfrak{g} + \xi_0\mathfrak{g}')(\xi_0) + O(\delta^2)$, and $-(\mathfrak{g} + \xi\mathfrak{g}') = \mathcal{H}_2$.

In particular, if ξ_0 remains bounded, the leading statement is

$$n\sqrt{\frac{n\log n}{2b}\varphi(\lambda)}\,R_b^n a_n = \mathfrak{g}\left(\frac{n(1-\zeta(2+\varepsilon)/b)}{\sqrt{n\log n\,\varphi(\lambda)/(2b)}}\right) + o(1). \tag{4.11}$$

At the critical choice $b = A = \zeta(2+\varepsilon)$,

$$a_n \sim R_b^{-n}\sqrt{\frac{A}{\pi\varphi(\lambda)}}\, n^{-3/2}(\log n)^{-1/2}, \tag{4.12}$$

uniformly over the bounded moving-order window. This is a joint transition law; letting λ tend to infinity is outside its stated uniform parameter set.

5 Lambert resummation and nonlinear universality

5.1 A convergent chart for the scale itself

The coefficient expansion is asymptotic. The scalar scale B has a stronger analytic structure that explains how it absorbs the $\log\log n$ corrections.

Proposition 5.1 (Convergent two-variable scale chart)**.** *Let $h = L^{-1}$, $w = L^{-1}\log L$, and $c = \log(\varphi(\lambda)/(2b))$. There is a holomorphic germ $Y(h, w; \lambda, b)$, with $Y(0, 0; \lambda, b) = 1$, uniform on compact real (λ, b) sets as above, such that*

$$B^2 = B_0^2 Y(L^{-1}, L^{-1}\log L; \lambda, b). \tag{5.1}$$

It is determined by the removable holomorphic equation

$$\begin{aligned} Y = {} & \frac{1 - e^{-\lambda}E_2(2\lambda h)e^{-\lambda w - \lambda h(c+\log Y)}}{\lambda\varphi(\lambda)} \\ & + \frac{2h}{\varphi(\lambda)}\left\{Z(2\lambda h) + \frac{\zeta(2+2\lambda h)^2}{b^2}\right\}. \end{aligned} \tag{5.2}$$

Its Taylor series in the two independent variables converges after physical substitution for sufficiently large L.

Proof. The quotient is holomorphic at $\lambda = 0$, since its numerator vanishes there. At $h = w = 0$, the right side equals $(1 - e^{-\lambda})/(\lambda\varphi(\lambda)) = 1$ and its Y derivative is zero. The implicit function theorem therefore applies with derivative one after moving Y to the left. A finite compactness cover and uniqueness give a common neighborhood. Substituting $B^2 = B_0^2 Y$ into (2.12) gives (5.2) exactly. The unique large positive solution from theorem 2.3 identifies the branch. □

5.2 Exact marginal Lambert scale

At $\varepsilon = 0$, equation (2.12) is

$$\frac{bB^2}{n} = \log B + \frac{3}{2} + \nu, \qquad \nu = \frac{\zeta(2)^2}{b^2}.$$

Set $T = 2bB^2/n$. Then

$$T - \log T = L + 3 + 2\nu - \log(2b), \tag{5.3}$$

$$T = -W_{-1}\left(-\frac{2b}{n}e^{-3-2\nu}\right), \qquad B^2 = \frac{n}{2b}T, \qquad q = \frac{2}{T}. \tag{5.4}$$

The lower real branch is required because $T \to \infty$ [9]. This is a scale for the coefficient law. The coefficient scale and the earlier inverse-function Lambert core have different constants; the following exact matching explains them.

Corollary 5.2 (All marginal central coefficients as constants)**.** *Let $s = 3$, $b = A = \zeta(2)$, and let $T > 1$ solve*

$$T - \log T = L + 5 - \log(2A). \tag{5.5}$$

Define

$$D_j = \frac{1}{\sqrt{\pi}\, j!} \operatorname{Re}\left[(\partial_z + \log 2 + i\pi)^j \Gamma(z)\right]_{z=j+1/2}, \qquad D_0 = 1. \tag{5.6}$$

For each fixed N,

$$R_b^n a_n = \sqrt{\frac{A}{\pi}}\, n^{-3/2} T^{-1/2} \left\{ \sum_{j=0}^{N} \frac{D_j}{T^j} + O(T^{-N-1}) \right\}. \tag{5.7}$$

If $Q_j(Y)$ denotes the Gamma-moment polynomial of [5], then exactly

$$D_j = Q_j(-\log 2). \tag{5.8}$$

Thus the exact Lambert scale turns the previous logarithmic polynomials into a sequence of constants.

Proof. At $\varepsilon = 0$ only $k = j$ survives in (3.4). Since $q = 2/T$, the normalized central coefficient of T^{-j} is $(-1)^j \mathcal{H}_{2j}^{[j]}(0)/(j!\mathfrak{g}(0))$. Equation (3.13) gives precisely (5.6). The leading prefactor is $\mathfrak{g}(0)/(nB) = \sqrt{A/\pi}\, n^{-3/2} T^{-1/2}$. Finally the predecessor defines

$$Q_j(Y) = \frac{1}{\sqrt{\pi}\, j!} \operatorname{Re}\left[(\partial_z - Y + i\pi)^j \Gamma(z)\right]_{z=j+1/2},$$

so substitution gives (5.8), with no asymptotic comparison needed. □

For $C = \gamma + \log 2$, the first constants are

$$\begin{aligned} D_1 &= 1 - \frac{C}{2}, \\ D_2 &= 1 - 2C + \frac{3C^2}{8} - \frac{3\pi^2}{16}. \end{aligned} \tag{5.9}$$

Re-expanding (5.5) in L gives, in particular,

$$R_b^n a_n = \sqrt{\frac{A}{\pi}}\, n^{-3/2} L^{-1/2} \left\{ 1 - \frac{\log L - \log A + \gamma + 3}{2L} + O\left(\frac{(\log L)^2}{L^2} \right) \right\}. \tag{5.10}$$

This agrees with the integrated marginal result, including its constant $\gamma + 3$. At higher orders the same conclusion follows from the exact identity (5.8) and the convergent scale chart, rather than from a comparison of a few numerical values.

5.3 Positive analytic outer functions

The square root is not essential to the all-order logarithmic calculus. Let G vary in a family of holomorphic functions satisfying the following uniform hypotheses for all sufficiently small real ε:

(i) $G(0) = 0$, and the Taylor coefficients of G at zero are nonnegative real numbers;

(ii) there are $\eta, M > 0$ such that G is holomorphic on $|w| < \zeta(3+\varepsilon) + \eta$ and bounded by M there;

(iii) $G'(0) \geq \eta$.

Shrinking η makes it harmless to use the same symbol for the uniform lower bound and the disk margin. Compact families with fixed positive margins satisfy these conditions. Put

$$f(z) = G(\mathrm{Li}_{3+\varepsilon}(z)), \quad c = G'(\zeta(3+\varepsilon)), \quad \kappa = \frac{A^2 G''(\zeta(3+\varepsilon))}{c}, \quad \mu = cA. \tag{5.11}$$

Then c is bounded above and away from zero, and κ is nonnegative and bounded. Define X_n using (2.5) and set

$$\frac{B^2}{nc} = K_\varepsilon(\log B) + \kappa =: V, \qquad q = \frac{E_2(\varepsilon) B^{-\varepsilon}}{V}, \qquad \xi_m = \frac{m - n\mu}{B}. \tag{5.12}$$

Theorem 5.3 (Two-derivative composition principle)**.** *Under the preceding uniform hypotheses, theorem 4.1 holds with (5.12) in place of (2.12)–(2.13), and with precisely the same functions $\mathcal{C}_j$. The leading scale is*

$$B^2 \sim \frac{ncL}{2}\varphi(\lambda). \tag{5.13}$$

For the germ $v(r) = f(re^{v(r)})$, write $a_n = [r^n]v(r)$ and $R = e^{-f(1)}$. Then $nBR^n a_n$ has the same expansion at $\xi = (n - n\mu)/B$.

After normalization by $f(1)$ and centering, all inverse-logarithmic orders depend on the outer function only through its first two derivatives at $\zeta(3+\varepsilon)$. Higher derivatives contribute to the remainder beyond every fixed inverse-logarithmic order.

Proof. The positive Taylor coefficients and the linear term of G imply $[z^k]f(z) \geq \eta k^{-3-\varepsilon}$. Uniform boundedness on the larger disk supplies uniform endpoint derivatives by Cauchy's estimates. Writing $D = \zeta(s) - \mathrm{Li}_s(e^{-t})$, Taylor expansion at the endpoint gives

$$G(\zeta(s) - D) - G(\zeta(s)) = -cD + \frac{1}{2}G''(\zeta(s))D^2 + O(|D|^3).$$

The proof of lemma 2.2 therefore yields

$$f(e^{-t}) - f(1) + \mu t = \frac{ct^2}{2}\{K_\varepsilon(-\operatorname{Log} t) + \kappa\} + O(|t|^{5/2}).$$

The scale existence proof applies with b replaced by $1/c$ and ν replaced by κ. The positive-intensity proof of lemma 4.2 also applies verbatim with the indicated uniform lower bound. The entire finite-expansion proof depends only on these two facts, so it gives the same $\mathcal{C}_j$ and remainder. Lagrange inversion proves the last coefficient identity exactly as in proposition 2.1. Terms involving higher derivatives of G begin at $O(D^3)$ and are covered by the uniform local remainder. □

For the square-root family, choose the disk margin smaller than $b_-^2/2$. Its branch point is at $w = u^2/2 = \zeta(s) + b^2/2$, and $G'(0) = 1/u$ is uniformly positive. Thus the main quadratic model satisfies the hypotheses, with uniform boundedness on a slightly smaller common margin.

Outer function $G(w)$	c	κ
aw	a	0
$u - \sqrt{u^2 - 2w}$	b^{-1}	A^2/b^2
$a(e^{\beta w} - 1)/\beta$	$ae^{\beta\zeta(s)}$	βA^2
$-(a/\beta)\log(1 - \beta w)$	$a/(1 - \beta\zeta(s))$	$\beta A^2/(1 - \beta\zeta(s))$

Here $a > 0$, $\beta > 0$, and in the last row $\beta\zeta(s) < 1$ with a uniform margin. The limit $\beta = 0$ recovers the linear row. Compact parameter subsets satisfying the stated margins provide concrete examples of the theorem.

The principle does not say that these models have the same full transseries. Their coefficients at exponential scales in L can depend on higher endpoint derivatives and on other singularities.

6 A growing positive window and its Lambert transition

The absolute error in Theorem 4.1 is uniform in the lattice location. Its usefulness is therefore not confined to bounded standardized displacements. We now use its explicit kernels to obtain a relative approximation in growing positive windows.

Throughout this section, put $L = \log n$ and retain the scale B, the parameters q, ε, and the centered coordinate $\xi = (m - nA/b)/B$ of Theorem 4.1. Write

$$\mathcal{P}_n(\xi) = B\mathbb{P}(X_n = m), \qquad g(\xi) = \mathfrak{g}(\xi) = \frac{\exp(-\xi^2/2)}{\sqrt{2\pi}}.$$

Only lattice coordinates are meant when $\mathcal{P}_n$ occurs. The uniform parameter hypotheses of that theorem give

$$\frac{c}{L} \le q \le \frac{C}{L}, \qquad |\varepsilon| \le \frac{C}{L}, \tag{6.1}$$

with positive constants independent of the parameters. In particular, the constants in the statements below are uniform for the scaled order in a fixed compact real set and for b in a compact subinterval of $(0, \infty)$.

6.1 Right tails of the explicit kernels

We use the notation

$$\mathcal{H}_{2k}^{[j]}(x) = \frac{\partial^j}{\partial\alpha^j}\mathcal{H}_\alpha(x)\bigg|_{\alpha=2k}, \qquad \mathcal{H}_\alpha(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-iyx - y^2/2}(-iy)^\alpha\, dy,$$

where $\operatorname{Re}\alpha > -1$ and the principal powers are used.

Lemma 6.1 (Kernel tails and order derivatives)**.** *For fixed integers $k \geq 1$ and $j \geq 1$, as $x \to +\infty$,*

$$\mathcal{H}_{2k}^{[j]}(x) = O(x^{-2k-1}(1+\log x)^{j-1}). \tag{6.2}$$

More precisely, if $a(\alpha) = 1/\Gamma(-\alpha)$, then

$$\mathcal{H}_{2k}^{[j]}(x) = x^{-2k-1}\sum_{r=1}^{j}\binom{j}{r}a^{(r)}(2k)(-\log x)^{j-r} + O(x^{-2k-11/4}). \tag{6.3}$$

In particular,

$$\mathcal{H}_{2}^{[1]}(x) = -2x^{-3}\left\{1 + 6x^{-2} + O(x^{-4})\right\}. \tag{6.4}$$

The constants can depend on k and j, which are fixed here.

Proof. We give a convolution proof so that differentiation with respect to the order does not rest on an unstated uniformity of a special-function asymptotic formula. Fix k, set $d = 2k+1$, and let $|\alpha - 2k| \leq 1/4$. Then

$$\mathcal{H}_{\alpha}(x) = \frac{1}{\Gamma(d-\alpha)}\int_0^{\infty} t^{d-\alpha-1}g^{(d)}(x-t)\,dt. \tag{6.5}$$

To verify the identity, first insert $e^{-\eta t}$ with $\eta > 0$ in the integral before differentiating d times in x. The Fourier multiplier of the resulting convolution is $(\eta - iy)^{\alpha-d}$. Differentiation multiplies it by $(-iy)^d$. As $\eta \downarrow 0$, this tends to $(-iy)^{\alpha}$. It is dominated, up to a constant uniform on the indicated order disc, by $|y|^{\operatorname{Re}\alpha}e^{-y^2/2}$: here $\operatorname{Re}(\alpha - d) < 0$ and $|\eta - iy| \geq |y|$. This is integrable. Dominated convergence on the physical side is also valid, because the power at $t = 0$ has real part greater than -1 and every derivative of g has Gaussian decay at infinity. This proves (6.5), jointly holomorphically in α on a neighborhood of the closed disc.

Put $p = d - \alpha - 1$ and change variables to $u = t - x$. On $|u| \leq x/2$, expand $(x+u)^p$ in powers of u/x through degree $d + 2M + 1$, for any fixed $M \geq 0$. The Taylor remainder is bounded by

$$C_{k,M}x^{\operatorname{Re}p-d-2M-2}|u|^{d+2M+2}$$

uniformly on the order disc. Integration against $|g^{(d)}(-u)|$ is harmless. Replacing the central integrals by integrals over the whole real line changes the answer by a quantity bounded by $e^{-x^2/16}$ times a fixed power of x. The part of the original integral where $|u| > x/2$ has the same type of bound. For the left part, the integrable factor $(x+u)^{\operatorname{Re}p}$ near $u = -x$ causes no difficulty; for the right part, polynomial growth is dominated by the Gaussian.

Integration by parts gives

$$\int_{\mathbb{R}} u^r g^{(d)}(-u)\,du = 0 \quad (r < d \text{ or } r - d \text{ odd}),$$

and, for $r = d + 2\ell$,

$$\int_{\mathbb{R}} u^{d+2\ell}g^{(d)}(-u)\,du = \frac{(d+2\ell)!}{2^{\ell}\ell!}.$$

It follows that, uniformly on the order disc,

$$\mathcal{H}_{\alpha}(x) = \sum_{\ell=0}^{M}\frac{x^{-\alpha-2\ell-1}}{2^{\ell}\ell!\,\Gamma(-\alpha-2\ell)} + O(x^{-\operatorname{Re}\alpha-2M-3}). \tag{6.6}$$

The exponentially small terms just described can be retained separately; they are already absorbed by this remainder.

The remainder in (6.6) is holomorphic in α. Cauchy's formula on the circle of radius $1/4$ about $2k$ bounds any fixed order derivative of that remainder by $O(x^{-2k-2M-11/4})$. With $M = 0$, differentiating the leading term gives (6.3): the $r = 0$ term vanishes because $a(2k) = 0$. This proves (6.2). For (6.4), use $M = 2$ and $k = j = 1$. The reciprocal Gamma function has a simple zero at each nonpositive integer, and

$$\frac{d}{d\alpha}\frac{1}{\Gamma(-\alpha - 2\ell)}\bigg|_{\alpha=2} = -(2 + 2\ell)!.$$

The first two terms are therefore $-2x^{-3}$ and $-12x^{-5}$; the remaining displayed term and the differentiated remainder are $O(x^{-7})$, as asserted. □

The same tail structure follows from the classical connection formula for parabolic cylinder functions and their large-variable expansions [8]: $\mathcal{H}_\alpha(x) = e^{-x^2/4}D_\alpha(-x)/\sqrt{2\pi}$,

$$D_\alpha(-x) = \cos(\pi\alpha)D_\alpha(x) + \frac{\pi}{\Gamma(-\alpha)}V(-\alpha - 1/2, x).$$

Here the first term contributes Gaussian decay and the second contributes the power tail. The convolution proof above supplies the parameter uniformity and the order derivatives directly.

6.2 A relative Gaussian and power-tail approximation

Theorem 6.2 (Positive windows of polynomial size in $\log n$)**.** *Fix $M > 0$, and let $r_L \to \infty$ with $r_L \le L^M$. Uniformly over lattice points and admissible parameters for which $r_L \le \xi \le L^M$,*

$$\mathcal{P}_n(\xi) = \left\{g(\xi) + q\xi^{-3}\right\}\{1 + o(1)\}. \tag{6.7}$$

Consequently, for each fixed $\delta \in (0, 2)$, whenever the stated intervals are nonempty,

$$\mathcal{P}_n(\xi) \sim g(\xi), \qquad r_L \le \xi \le \sqrt{(2 - \delta)\log L}, \tag{6.8}$$

$$\mathcal{P}_n(\xi) \sim q\xi^{-3}, \qquad \sqrt{(2 + \delta)\log L} \le \xi \le L^M. \tag{6.9}$$

No relative conclusion about negative growing displacements is included in this theorem.

Proof. Choose an integer $N > 3M$ in Theorem 4.1. Its term of total degree one is $-q\mathcal{H}_2^{[1]}(\xi)/2$. By (6.4), this equals $q\xi^{-3}\{1 + O(\xi^{-2})\}$. For every term with total degree $j \ge 2$, divide its absolute value by $q\xi^{-3}$. Equations (6.1) and (6.2) bound the quotient by a constant times

$$q^{k-1}|\varepsilon|^{j-k}\xi^{-2k+2}(1 + \log\xi)^{j-1} \le C_N\left(\frac{1 + \log\xi}{L}\right)^{j-1}.$$

Only finitely many terms occur. Their sum is $o(1)$ uniformly because $\log\xi \le M\log L$. The absolute remainder divided by $q\xi^{-3}$ is

$$O(\xi^3 L^{-N}) = O(L^{3M-N}) = o(1).$$

Finally $\xi^{-2} \le r_L^{-2} \to 0$. We have thus proved the stronger additive statement $\mathcal{P}_n(\xi) = g(\xi) + q\xi^{-3}\{1 + o(1)\}$, which implies (6.7) since both main terms are positive.

For (6.8), the ratio of the power term to the Gaussian is at most a constant times $L^{-\delta/2} r_L^{-3}$. For (6.9), the ratio of the Gaussian to the power term is at most a constant times $L^{-\delta/2}(\log L)^{3/2}$: the function $x^3 e^{-x^2/2}$ decreases for $x > \sqrt{3}$. Both bounds tend to zero, proving the two consequences. □

In the coefficient specialization of Theorem 4.1, $m = n$, and hence the same conclusions apply directly to $nBR_b^n a_n$. Coordinates $\xi \le L^M$ force $b \to A$ because $1 - A/b = \xi B/n = O(L^{M+1/2}/\sqrt{n})$. Thus these are growing positive windows adjacent to the feedback threshold. They are distinct from a fixed supercritical parameter regime, whose standardized displacement is much larger.

Corollary 6.3 (Relative matching with the supercritical coefficient law)**.** *In the diagonal specialization, uniformly for each fixed $M > 0$ and $\delta > 0$ in the nonempty range*

$$\sqrt{(2+\delta)\log L} \le \xi \le L^M,$$

one has

$$R_b^n a_n \sim \frac{1}{b\{n(1 - A/b)\}^{3+\varepsilon}} = \frac{b^{s-1}}{(b-A)^s}\, n^{-s}. \tag{6.10}$$

Proof. On this window $E_2(\varepsilon) \to 1$ and $\xi^{-\varepsilon} \to 1$ uniformly, since $|\varepsilon \log \xi| = O(\log L/L)$. Thus the power term in (6.9) is relatively equivalent to $q\xi^{-3-\varepsilon}/E_2(\varepsilon)$. The exact identities

$$\frac{q}{E_2(\varepsilon)} = \frac{n}{bB^{2+\varepsilon}}, \qquad B\xi = n(1 - A/b)$$

give (6.10) after division by nB. □

There is an exact explanation for the amplitude cancellation. The first divided-difference correction is $-q(\mathcal{H}_{2+\varepsilon} - \mathcal{H}_2)/(2\varepsilon)$. Its leading algebraic tail is $q\xi^{-3-\varepsilon}/E_2(\varepsilon)$ because $1/\Gamma(-2-\varepsilon) = -2\varepsilon/E_2(\varepsilon)$. The polynomial-log window makes this term relatively equivalent to the simpler $q\xi^{-3}$ while preserving the original coefficient amplitude in (6.10).

6.3 An exact Lambert location and a two-term profile

Theorem 6.4 (Lambert location of the Gaussian–power transition)**.** *For sufficiently small $q > 0$, let $x_*(q) > \sqrt{3}$ be the larger positive solution of $g(x_*) = qx_*^{-3}$. It has the exact formula*

$$x_*(q)^2 = -3W_{-1}\left(-\frac{(2\pi)^{1/3}}{3} q^{2/3}\right). \tag{6.11}$$

If $Q = \log(1/q)$, then

$$x_*(q)^2 = 2Q + 3\log(2Q) - \log(2\pi) + O\left(\frac{\log Q}{Q}\right). \tag{6.12}$$

For the parameters in (6.1), uniformly for τ in compact real intervals and positive lattice coordinates $\xi_n > 0$ satisfying

$$\xi_n^2 = x_*(q)^2 + 2\tau + o(1), \tag{6.13}$$

where the final error is uniform, one has

$$\frac{\mathcal{P}_n(\xi_n)}{qx_*(q)^{-3}} \longrightarrow 1 + e^{-\tau}. \tag{6.14}$$

In particular $x_*(q) \sim \sqrt{2\log\log n}$.

Proof. The function $x^3 g(x)$ increases up to $\sqrt{3}$ and decreases thereafter, so the stated larger root exists and is unique for small q. The defining equation is equivalent to

$$x_*^2 e^{-x_*^2/3} = (2\pi)^{1/3} q^{2/3},$$

which gives (6.11) with the branch W_{-1} [9]. Alternatively, taking logarithms gives the exact scalar equation

$$y = 2Q + 3\log y - \log(2\pi), \qquad y = x_*^2. \tag{6.15}$$

It first implies $y \sim 2Q$. Substituting $y_0 = 2Q + 3\log(2Q) - \log(2\pi)$ into its two sides leaves a residual $O((\log Q)/Q)$, while the derivative of $y - 3\log y$ is $1 - 3/y$, bounded away from zero at the larger root. The mean value theorem proves (6.12).

By (6.1), $Q = \log L + O(1)$. Therefore (6.13) lies in the window of Theorem 6.2 for any fixed $M > 0$ and a suitable $r_L \to \infty$. Moreover,

$$\frac{g(\xi_n)}{qx_*(q)^{-3}} = \frac{g(\xi_n)}{g(x_*(q))} = e^{-\tau + o(1)}, \qquad \frac{q\xi_n^{-3}}{qx_*(q)^{-3}} = \left(\frac{x_*(q)}{\xi_n}\right)^3 \longrightarrow 1.$$

The relative mixture theorem now gives (6.14). □

The lattice restriction is immaterial to the transition coordinate: rounding $nA/b + B\sqrt{x_*(q)^2 + 2\tau}$ to the nearest integer changes ξ_n^2 by $O(x_*(q)/B) = o(1)$. For diagonal coefficients, the feedback parameter can instead be varied on the locally unique branch near $b = A$, choosing the compact feedback interval to contain $\zeta(2)$ in its interior. Here is an explicit verification of the latter assertion. Hold n, s fixed, put $V = bB^2/n$, and differentiate the defining scale equation. It gives

$$\beta := \partial_b \log B = -\frac{b^{-1} + 2A^2/(b^3 V)}{2 - q} = O(1), \qquad \partial_b \log q = -b^{-1} - (2+\varepsilon)\beta = O(1).$$

For $\xi(b) = n(1 - A/b)/B$, consequently,

$$\xi'(b) = \frac{nA}{b^2 B} - \xi(b)\beta \asymp \sqrt{n/L} > 0$$

through any window $\xi = O(L^M)$ near $b = A$. The derivative of $x_*(q(b))$ is only $O(1/x_*)$, by (6.15). Starting from $\xi(A) = 0$, continuity and these derivative bounds yield a unique local feedback choice for each bounded τ in (6.13).

6.4 Relation to established probability results

The Gaussian and power-tail competition is an established phenomenon. Lam, Blanchet, Burch, and Bazant obtain nonclassical corrections to Gaussian limits for regularly varying densities, express corrections using Dawson and parabolic cylinder functions, and explain their connection with heavy-tail large deviations [10]. Berger, Birkner, and Yuan treat large and local large deviations,

including the infinite-variance normal domain with tail index -2, and identify the transition between collective Gaussian behavior and a single exceptionally large summand [11].

Accordingly, neither the existence of such a competition nor the use of parabolic cylinder functions is claimed as new here. The result established for the present model is the uniform extension through the varying order $s_n = 3 + O(1/\log n)$ and varying nonlinear feedback, with the explicit paired normalization and the finite all-order Stirling formulas of Theorem 4.1. The growing windows above follow from its uniform absolute remainder, and the exact Lambert coordinate (6.11) describes their Gaussian–power transition in that normalization. The fixed-law results just cited do not, without additional uniform estimates, supply this triangular-parameter assertion.

7 A local classification at reciprocal orders

This section sharpens the nonintegral inverse chart and establishes an obstruction to the extension proposed in Research Question 9 of *Resonant Order Jets and Convergent Inverse Transseries: for Polylogarithms and Logarithmically Weighted Tails* [6]. The starting expansion is classical; the classification and uniform reciprocal-order chart below are deductions from it. The source theorem is not contradicted: its global endpoint-circle assertion is stated only for integer orders.

The nonintegral charts in the predecessor imply an exceptional phenomenon that is worth separating from the integer-order resonance theory. Write $s = 1 + \alpha > 1$ and

$$D_\alpha(t) = \zeta(1+\alpha) - \mathrm{Li}_{1+\alpha}(e^{-t}).$$

For nonintegral α, the classical convergent expansion [7] gives

$$D_\alpha(t) = b(\alpha)t^\alpha + tA_\alpha(t), \qquad b(\alpha) = -\Gamma(-\alpha), \qquad A_\alpha(t) = \sum_{k\geq 1} \frac{(-1)^{k+1}\zeta(1+\alpha-k)}{k!}t^{k-1}.$$

Initially $0 < |t| < 2\pi$, $-\pi < \arg t < \pi$, and $t^\alpha = \exp(\alpha \log t)$ uses the principal logarithm. Further continuation is on the logarithmic covering. The function $A_\alpha(t)$ is jointly holomorphic for t near zero and α in any sufficiently small neighborhood avoiding the nonnegative integers. Let $t_\alpha(y)$ denote the inverse which is positive for positive small y.

Theorem 7.1 (Minimal local ramification)**.** *For real $s > 1$ the minimal degree of a finite ramification $y = w^N$ making the endpoint inverse $t_{s-1}(y)$ holomorphic is as follows.*

1. *If $0 < \alpha = s - 1 < 1$ and $\alpha = p/q$ is in lowest terms, the minimal degree is p.*
2. *If $\alpha > 1$ is nonintegral and $\alpha = p/q$ is in lowest terms, the minimal degree is q.*
3. *If α is irrational, or if s is a positive integer at least two, no finite ramification suffices.*

In particular, the endpoint inverse is holomorphic without ramification if and only if $s = 1 + 1/m$ for an integer $m \geq 2$. The same classification holds for the endpoint germ $\mathrm{Li}_s^{\langle -1\rangle}(\zeta(s) - y) = e^{-t_{s-1}(y)}$.

Proof. Suppose first that $0 < \alpha = p/q < 1$. The substitution $t = \xi^q$ makes the forward germ holomorphic:

$$D_\alpha(\xi^q) = b\xi^p + \xi^q A_\alpha(\xi^q).$$

The factor multiplying ξ^p is nonzero at zero, so an analytic change of coordinate reduces this germ to η^p. Its local inverse is therefore holomorphic in $y^{1/p}$, and its t coordinate starts with $b^{-q/p}y^{q/p}$.

Since p and q are coprime, this leading term forces every possible ramification degree to be divisible by p. Thus the minimal degree is exactly p.

If $\alpha = p/q > 1$ is nonintegral, the same substitution gives a forward germ of order q, because $A_\alpha(0) = \zeta(\alpha) > 0$. It follows that the inverse t is holomorphic in $y^{1/q}$. To see minimality, let $T(y)$ be the analytic inverse of $tA_\alpha(t)$. Expansion of the inverse equation gives

$$t_\alpha(y) = T(y) - b\zeta(\alpha)^{-\alpha-1}y^\alpha + o(y^\alpha),$$

after retaining the analytic terms of T through the relevant degree. The displayed fractional coefficient is nonzero, so q divides every possible ramification degree.

For irrational $\alpha < 1$, the leading exponent $1/\alpha$ is irrational. For irrational $\alpha > 1$, the preceding first nonanalytic exponent is α. Neither can occur in a convergent Puiseux germ. At $s = 2$, $D_1(t) = t(1 - \log t) + O(t^2)$, whence $t_1(y) \sim y/\log(1/y)$, which is not Puiseux. At integer $s = m + 1 \geq 3$, the singular term of the forward deficit is $(-1)^m t^m \log t/m!$, and its linear coefficient is $a = \zeta(m) > 0$. Therefore the inverse has a nonzero logarithmic term $(-1)^{m+1}y^m \log y/(m!a^{m+1})$ in addition to its analytic part. Finite ramification multiplies the logarithm by a nonzero integer and does not remove it. Finally $t \mapsto e^{-t}$ is biholomorphic near zero, so it preserves these ramification degrees. □

7.1 Explicit coefficients at the removable endpoints

Fix an integer $m \geq 2$, put $\alpha = 1/m$ and $b_m = -\Gamma(-1/m) > 0$, and define

$$G_m(\xi) = b_m + \sum_{k\geq 1} \frac{(-1)^{k+1}\zeta(1 + 1/m - k)}{k!}\xi^{mk-1}.$$

Then $y = \xi G_m(\xi)$, $t = \xi^m$, and Lagrange inversion gives the exact finite extraction formulas

$$[y^n]t_{1/m}(y) = \frac{m}{n}[\xi^{n-m}]G_m(\xi)^{-n},$$

$$[y^n]e^{-t_{1/m}(y)} = -\frac{m}{n}[\xi^{n-m}]e^{-\xi^m}G_m(\xi)^{-n} \qquad (n \geq 1).$$

Coefficients with a negative extraction exponent vanish. In particular

$$t_{1/m}(y) = b_m^{-m}y^m - m\zeta(1/m)b_m^{-2m}y^{2m-1} + O(y^{3m-2}).$$

This last remainder follows because the next possible degree in $t(y)$ is $3m - 2$ (the term with $k = 2$ first contributes at $3m - 1$). For the ordinary inverse endpoint germ, every derivative of order $1, \ldots, m - 1$ vanishes, and its mth derivative with respect to $w = \zeta(s) - y$ equals $(-1)^{m+1}m!b_m^{-m}$.

For example, at $s = 3/2$ one has $b_2 = 2\sqrt{\pi}$ and

$$t_{1/2}(y) = \frac{y^2}{4\pi} - \frac{\zeta(1/2)}{8\pi^2}y^3 + \frac{5\zeta(1/2)^2}{64\pi^3}y^4 + O(y^5).$$

Thus the square-root singularity of the forward polylogarithm has become a quadratic stationary point of its holomorphic endpoint inverse.

7.2 A uniform chart crossing a removable endpoint

The apparent disappearance of branching is not uniform under order perturbation. Nevertheless a short exact equation gives a convergent chart that is uniform at the reciprocal order.

Theorem 7.2 (Reciprocal-order crossover chart)**.** *Let $m \geq 2$ and let K be a compact real interval. Put*

$$\alpha = \frac{1}{m} + \epsilon, \quad T = \log(1/y), \quad \tau = \epsilon T, \quad h = T^{-1}, \quad r = y^{m-1}, \quad q = y^m.$$

There is a function $U(h, r, q; \tau)$ holomorphic in its first three variables near zero, with a complex neighborhood of K for the last variable and neighborhoods uniform on K, such that the physical inverse satisfies

$$t_\alpha(y) = y^m U(T^{-1}, y^{m-1}, y^m; \epsilon T).$$

Its value at the origin is

$$U(0, 0, 0; \tau) = b_m^{-m} e^{m^2 \tau}.$$

The Taylor series in the three independent small variables converges after physical substitution for sufficiently small positive y.

Proof. Substitute $t = y^m U$ into $D_\alpha(t) = y$ and divide by y. The exact equation is

$$b(1/m + h\tau) e^{-m\tau} U^{1/m + h\tau} + rU A_{1/m + h\tau}(qU) = 1.$$

For $h = r = q = 0$ its indicated solution is $U_0(\tau) = b_m^{-m} e^{m^2 \tau}$. Choose the logarithm $\log U_0(\tau) = -m \log b_m + m^2 \tau$ and write $U = U_0 V$. In this coordinate the equation is holomorphic near $V = 1$, and its V derivative at $h = r = q = 0$, $V = 1$ is $1/m$. The holomorphic implicit function theorem and a finite compactness cover of K give the claimed uniform chart. Local uniqueness glues the charts. The physical positive branch is selected by continuity from positive y and the strictly increasing real forward deficit. □

For each fixed $j \geq 1$, differentiation in α at fixed deficit y gives

$$\partial_\alpha^j t_\alpha(y)\Big|_{\alpha = 1/m} = b_m^{-m} (-m^2)^j y^m (\log y)^j + O(y^m (1 + |\log y|)^{j-1}).$$

One may prove this either from the uniform chart or by differentiating $b(\alpha)^{-1/\alpha} y^{1/\alpha}$ and the jointly holomorphic two-generator remainder. Derivatives of the latter acquire at most fixed powers of $\log y$ and retain a positive additional power of y. Thus order differentiation exposes an infinite logarithmic monodromy which is absent from the base germ. Here the independent variable is the deficit y; differentiation at fixed w also differentiates $\zeta(s) - w$ and is a different operation.

7.3 Why the integer endpoint-circle proof cannot extend verbatim

For $0 < \alpha = s - 1 < 1/2$ and $u \downarrow 0$, the upper-bank value is

$$\mathrm{Li}_{1+\alpha}(e^u + i0) = \zeta(1 + \alpha) + \Gamma(-\alpha) e^{-i\pi\alpha} u^\alpha + O(u).$$

Since $\Gamma(-\alpha) < 0$ and $\cos(\pi\alpha) > 0$, this yields

$$|\mathrm{Li}_s(e^u + i0)| = \zeta(s) + \Gamma(1 - s) \cos(\pi(s - 1)) u^{s-1} + o(u^{s-1}) < \zeta(s)$$

for every sufficiently small positive u. Consequently the cut-modulus assertion in the integer-order theorem fails for every $1 < s < 3/2$. The predecessor states that assertion only for integers, so this is a negative answer to a proposed extension, not a counterexample to a printed theorem.

The cut entering the endpoint circle does not itself determine the Taylor radius of an analytically continued inverse: the cut is a boundary of the chosen forward sheet, not necessarily an inverse singularity. Likewise endpoint removability does not rule out another singularity of the same modulus. Determining the global radius at the reciprocal orders remains a separate problem.

8 Verification, proposed repository updates, and further research

8.1 Independent finite computations

The proofs above do not depend on numerical evidence. The accompanying computations serve a different purpose: they check branch conventions, normalizations, finite coefficient identities, and the practical size of the first corrections. All programs are standalone and write their results into the package's `data` directory. The complete parameter grid and signed errors are retained, so the table below is not a selection of favorable cases.

For an independent finite computation of the coefficient, expand $f(z) = \sum_{k\ge1} c_k z^k$ and the probability generating function as $\sum_{k\ge0} p_k z^k$. The defining equations give

$$c_k = \frac{1}{u}\left\{ k^{-s} + \frac{1}{2}\sum_{j=1}^{k-1} c_j c_{k-j} \right\}, \tag{8.1}$$

$$p_0 = e^{-nv_c}, \qquad p_k = \frac{n}{k}\sum_{j=1}^{k} j c_j p_{k-j}, \qquad R_b^n a_n = \frac{p_n}{n}. \tag{8.2}$$

These finite positive recurrences are independent of the asymptotic correction formula. Their use is consistent with the predecessor's coefficient verification [1]. An additional calculation in exact rational arithmetic, at $s = 3, u = 4$, checks the first eight coefficients of the original implicit equation against the Lagrange representation, using two separate extraction routes.

The numerical grid takes

$$n \in \{128, 512, 2048, 4096\}, \qquad \lambda \in \{-1, 0, 1\}, \qquad \xi \in \{-1, 0, 1\}.$$

For each of the 36 cases the scale equation and the feedback parameter are solved simultaneously so that $\xi = n(1 - A/b)/B$ has the stated value. The approximation of order N means $\sum_{j=0}^{N} \mathcal{C}_j(\xi; q, \varepsilon)$, with the exact scale, compared with Bp_n. Table 1 reports the largest absolute relative error among the nine cases at each n. The finite values of n are diagnostics, rather than a claim that the eventual bounds in the proof have already reached their asymptotic constants.

The finite recurrences use extended floating-point arithmetic, with rescaling available when required. Four selected cases are repeated with 75-digit arithmetic; the largest relative discrepancy is below 2.7×10^{-17}. Repeating those reference calculations at 60 digits agrees to better than 4×10^{-60}. These comparisons check the particular computed values; they are not interval arithmetic certificates. The JSON output records the actual precision and platform. Some individual cases become slightly less accurate after another correction is added. No monotonicity of finite asymptotic truncations is asserted.

Table 1: Maximum absolute relative error, in percent, on the specified nine-case grid at each n. Order zero is the Gaussian term alone.

n	$N=0$	$N=1$	$N=2$	$N=3$
128	100.338	17.736	9.558	6.263
512	70.451	11.565	5.148	2.906
2048	54.720	8.177	3.049	1.466
4096	49.344	7.033	2.418	1.070

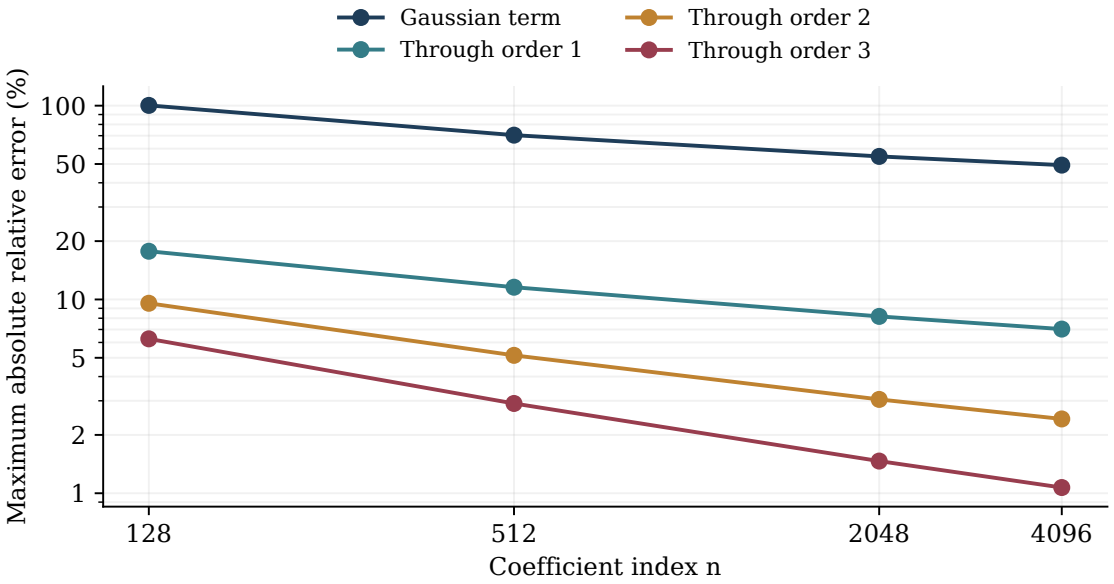


Figure 1: The errors in Table 1, computed from finite positive recurrences. The lines connect the four evaluated values of n and do not represent fitted convergence rates.

The symbolic program verifies six total-degree identities through degree five in the exponential–Stirling expansion; 36 central Gamma-jet identities; and the five identities $D_j = Q_j(-\log 2)$ for $0 \le j \le 4$. The two algebraic routes use, respectively, reciprocal Gamma products and positive-half-integer Gamma moments. Nine noninteger-order Fourier quadratures, at 80 digits, check the parabolic cylinder normalization with absolute errors below 10^{-65}. Separate Fourier checks cover the order derivatives used in the coefficient diagnostics. These computations are sensitive to the Fourier sign and to the imaginary part of $\mathrm{Log}(-iy)$.

For the inverse endpoint results, nine high-precision inverse calculations cover $m = 2, 3, 4$ and three small deficits y; six upper-bank evaluations cover $s = 1.25, 1.4$ and three small u. They agree with the proved reciprocal expansions and the negative cut-modulus increment. The exact classifications in Theorem 7.1 follow from analytic continuation and the displayed expansions, not from those samples.

Figure 2 illustrates only the normalized two-term expression in Theorem 6.4. This distinction matters: the figure is an illustration of the explicit analytic profile, whereas Figure 1 compares independently computed coefficients with their approximations.

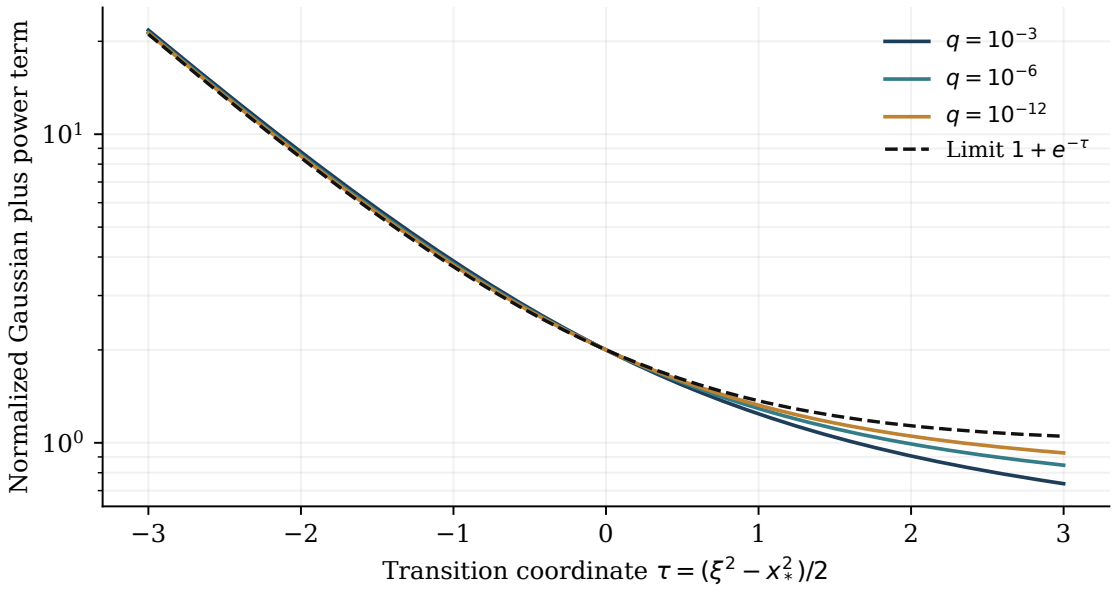


Figure 2: For $\xi = \sqrt{x_*(q)^2 + 2\tau}$, the exact normalized two-term expression is $e^{-\tau} + (1 + 2\tau/x_*(q)^2)^{-3/2}$. The curves show this expression for three values of q and its limit $1 + e^{-\tau}$. They are not sampled probabilities.

8.2 A focused audit and concrete proposed amendments

No false printed theorem was established in the coefficient sources used here. The useful corrections are statements of scope and updates to explicit research questions. The accompanying `INTEGRATION.md` gives proposed insertions without changing the pinned repository. `THEOREM_LEDGER.md` records the hypotheses and dependencies of each principal result. Source paths and hashes are supplied in `PROVENANCE.json`.

A1. Mark the compact joint window of Q1 as resolved. Theorem 4.1 proves the requested joint coefficient law, all fixed logarithmic corrections, and the uniform Fourier bound. Equation (4.10) supplies the first correction in the elementary scale requested there. Equations (5.3)–(5.8) give the exact marginal comparison. This is stronger than substituting $s = s_n$ into a theorem whose constants were controlled only at fixed s.

A2. Record a partial positive answer to Q2. Theorems 6.2 and 6.4 give a relative approximation on every fixed polynomial window in $\log n$, and an explicit transition profile. The negative growing window and the passage to genuinely macroscopic deviations remain open in this treatment.

A3. Retain the distinction between an endpoint value and the coefficient radius. The identity $R_b^n a_n = \mathbb{P}(X_n = n)/n$ holds for every $b > 0$. The reference $R_b = e^{-(u-b)}$ is the actual radius on the endpoint-controlled side $b \geq A$. For $b < A$, the principal inverse encounters an interior fold and the radius is larger. The uniform lattice estimate uses the reference value without relabeling it as the radius below criticality.

A4. Add the reciprocal-order exception to endpoint discussions. The inverse endpoint is holomorphic at $s = 1 + 1/m$, $m \geq 2$. An argument which treats every noninteger polylogarithm

branch point as a nonremovable inverse branch point misses these exceptions. Theorem 7.1 gives the complete local replacement, with the minimal ramification index. It makes no claim about other inverse singularities at the same modulus.

A5. Qualify the proposed real-order extension of Q9. The cut-modulus inequality used for integer orders fails for every $1 < s < 3/2$. Thus that proof cannot establish a blanket real-order extension of the endpoint-circle statement in [6]. This is a counterexample to the proposed extension of the inequality, not to the existing integer theorem, and it does not itself settle the global inverse Taylor radius.

A6. Keep the two convergence assertions separate. The implicit scale chart and the reciprocal-order inverse chart are convergent holomorphic expansions. The coefficient series is proved to every *fixed* logarithmic order, with constants depending on that order. Neither a convergent infinite Fourier core expansion nor an optimal-truncation theorem is proved. Uniform absolute error over all lattice points is also distinct from a relative approximation where the probability vanishes. The positive-window proof supplies the additional estimate in exactly the range where it is used.

8.3 Further research questions with concrete starting points

The following questions separate direct extensions from problems requiring a different analytic mechanism. None is needed for the validity of the results above.

R1. What is the relative law on growing negative windows? The Fourier theorem supplies an absolute error there, but the algebraic positive tail is absent. A saddle displacement on the negative side should be combined with the pole-paired exponent before truncation. A useful first target is a relative theorem for $-L^M \le \xi \le -r_L$, identifying the largest M for which a finite expansion in the same parameters suffices. This would address the remaining side of Q2 and quantify the onset of the interior-fold regime.

R2. Can one join the compact resonance window to the stable and finite-variance regions at every correction order? Here $|\lambda|$ is bounded. Letting λ tend to either infinity in the displayed formulas is not justified by this proof. The leading all-rate results in [3] suggest an overlap problem: construct uniform normalizations and remainders when $\lambda = \lambda_n$ grows, then match to fixed-order stable or finite-variance expansions. The exact function K_ε is a natural common coordinate, but the Fourier truncation and error constants must be redesigned.

R3. What is the first sector beyond all logarithmic orders, and how does it remember the nonlinearity? The composition theorem removes all endpoint derivatives of G beyond the second from the fixed logarithmic sector. Retain the cubic terms in $G(\zeta(s) - D_s(t))$ and scale them by nB^{-3}. This gives a concrete route to the first smaller sector, including the first dependence on $G'''(\zeta(s))$. The question is to prove a uniform mixed expansion in powers of B^{-1} and inverse logarithms, with a transparent description of which endpoint derivatives first occur in each sector.

R4. How fast do the coefficient functions grow with the truncation order? The finite formulas reduce this to joint large-order estimates for Stirling numbers and $\mathcal{H}_{2k}^{[j]}$. Determine a useful Gevrey bound on compact spatial sets, investigate optimal truncation, and identify the associated exponentially small remainder. Any summation procedure must respect the Fourier

tail: the untruncated exponential obtained from the local model need not be integrable on the full real line. Formal exponentiation alone therefore cannot supply this result.

R5. How far can the relative positive window be extended? The present argument reaches $\xi \le L^M$ for every fixed M, by choosing a sufficiently large but fixed logarithmic order. For displacements of algebraic size in n, this argument loses relative control. A direct uniform local large-deviation estimate for the triangular compound Poisson family could bridge the gap to fixed supercritical $b > A$, while keeping the exact amplitude $b^{s-1}(b-A)^{-s}$. The work of [11] is an appropriate starting point for the required probability estimates, with the moving-law uniformity proved explicitly.

R6. What determines the global inverse radius at a removable reciprocal endpoint? Local removal of the value $\zeta(1+1/m)$ does not locate the next singularity of the inverse germ. Determine which critical values, asymptotic values, or sheet obstructions first meet its continuation. Potential critical points away from zero satisfy $\mathrm{Li}_{1/m}(z) = 0$. Even the case $m = 2$ would clarify whether other singularities can retain the endpoint modulus after the positive endpoint itself becomes removable.

R7. Can order jets and ramification be organized uniformly in the jet index? The convergent reciprocal chart handles every fixed jet and compact τ. The leading term shows that differentiation in order restores logarithms despite local removability at the central order. Determine the allowable joint growth of the derivative index, $|\log y|$, and the winding number. This would connect the exact monodromy and order-jet calculus of the existing Polylogarithms chapters with quantitative inverse estimates.

R8. What is the sharp positivity hypothesis for composition universality? The sufficient condition $G'(0) > 0$ gives a uniform coefficient lower bound and aperiodicity immediately. Some positive outer functions with $G'(0) = 0$ may still have the needed tail mass and lattice span after composition. Characterize the weakest conditions that yield the two Fourier bounds, and then separate these probabilistic conditions from the local analytic conditions at $\zeta(s)$.

R9. Can several colliding tail terms be treated by one finite calculus? Replace a single polylogarithm by a positive combination with orders moving through the same integer, or add controlled slowly varying factors obtained by order differentiation. The scalar divided difference suggests a multivariate Stirling or Bell-polynomial description. The challenge is to choose a normalization that preserves uniform positivity and keeps all pole cancellations explicit.

R10. Which part should be formalized first? A tractable initial layer consists of the finite coefficient–probability identity, the Gamma recurrence pairing, the Stirling coefficient formula, and the reciprocal-order Lagrange identities. The analytic layer then needs parameter-uniform Fourier inversion, dominated convergence with logarithmic powers, and holomorphic implicit inversion. This separation would produce reusable formal components rather than a single opaque formalization of the final theorem.

The central structural point is that the collision at $s = 3$ can be absorbed into an exact scale while leaving a finite order-derivative calculus for every correction. Positivity provides the global Fourier control needed to turn that local calculus into a theorem. The reciprocal inverse endpoints exhibit the complementary phenomenon: ramification can disappear at an exceptional order, while its logarithmic dependence remains visible to differentiation in order.

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